01 SEED Seed Initialization

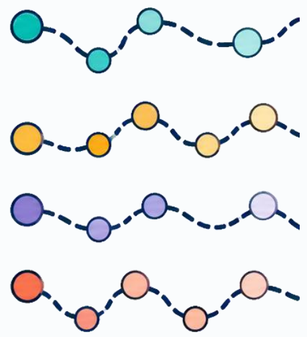
Start with rare, topology-rich entities across domains



Rare-entity pool
Chinese + English Wikipedia



Topology probe
multi-start random walks



Coverage test
deep · wide · locally diverse



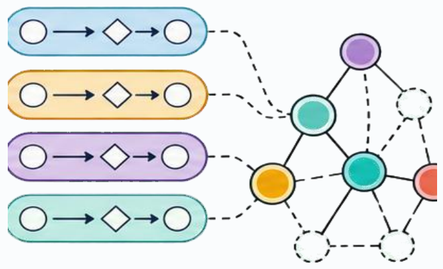
Domain-balanced seeds
prior labels + incremental update

02 WEB Web-enhanced Agentic Expansion

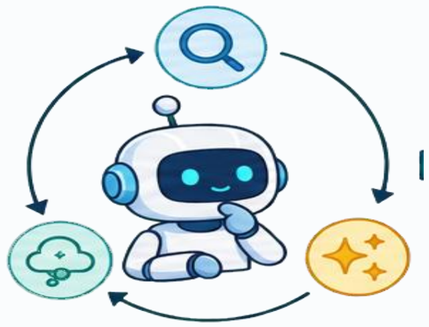
Search, extract, locate gaps and iteratively expand



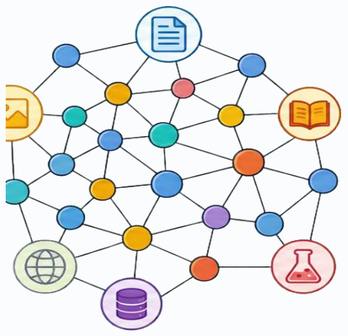
Agentic web queries
ask beyond internal knowledge



Long-tail fact extraction
web pages → relation triples



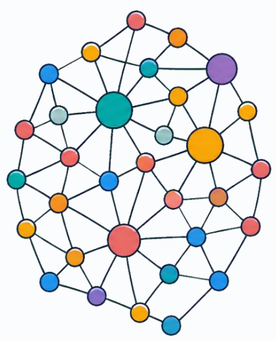
Missing-link expansion
locate gaps · query again



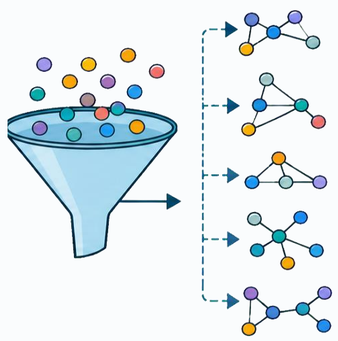
Global graph fusion
disambiguate · aggregate · iterate

03 SAMPLE Knowledge Sampling + Difficulty Ranking

Generate variants, sample sub-graphs, rank difficulty



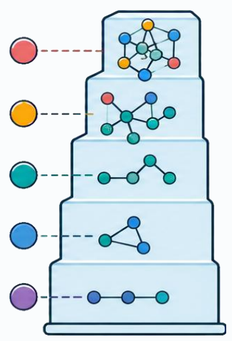
Bulk subgraph sampling
nodes · depth · width · branching



Candidate subgraphs
large-batch structural variants



Difficulty factors
degree · depth · width · source



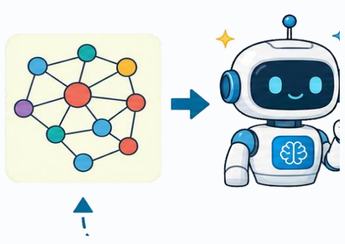
Difficulty ranking
custom composite score



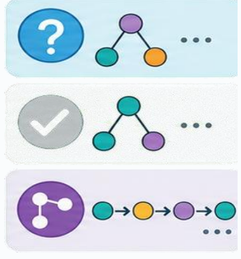
Retain hard subgraphs
high-complexity QA context

04 SYNTHESIZE QA Generation + Complexity Scaling

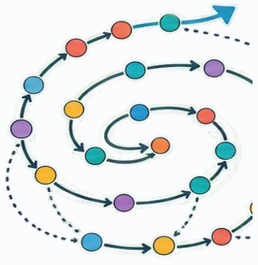
Scale reasoning depth, tool use, trajectory length, and style



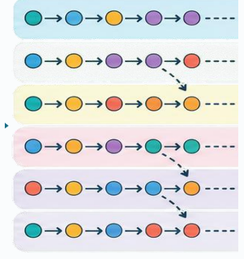
Strong-LLM synthesis
hard subgraph as context



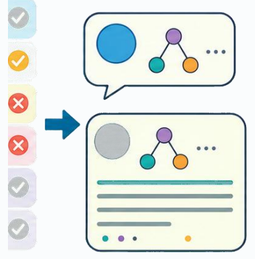
Initial QA triples
Q · A · factual reasoning chain



Complexity scaling
more hops · deep / wide search



Pass@K trajectory filter
select long tool-use traces



Multi-style QA outputs
multi-hop · dialogue · long-form

Hard, long-trajectory examples feedback into synthesis